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AN OPTIMIZED REINFORCEMENT LEARNING-BASED ROUTING PROTOCOL USING SPIDER MONKEY OPTIMIZATION ALGORITHM

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Abstract. In Wireless Sensor Networks, numerous machine-learning algorithms are used in the packet routing strategy. The recommended routing approach is implemented using adaptive reinforcement learning and a spider monkey optimization algorithm. Data packets are sent over the network via optimum route establishment—the above attempts to save energy by changing the Q-value at each instance. Technology enables adaptive mechanisms to demonstrate intelligent behavior in complex and dynamic situations such as Wireless Sensor Networks (WSN). Recently, new routing methods for WSNs have been devised in response to such a notion. Due to user interference and overall network performance, the numerous routing pathways cause frequent channel-switching occurrences. The spider monkey optimization method obtains routing metrics such as bandwidth, end-to-end latency, throughput, overhead, and packet delivery ratio. Compared to other machine learning algorithms, such as RLBR, the performance of WSN is improved and obtained better performance results.

Keywords: Reinforcement Learning; Spider Monkey Optimization algorithm, Multi-hop routing; Wireless Sensor Networks.

1. Introduction

The WSN is also widely recognized as a network without infrastructure. A wireless network consists of multiple nodes designed to track environmental factors. A WSN is meant for communicating and exchanging data between sensors.



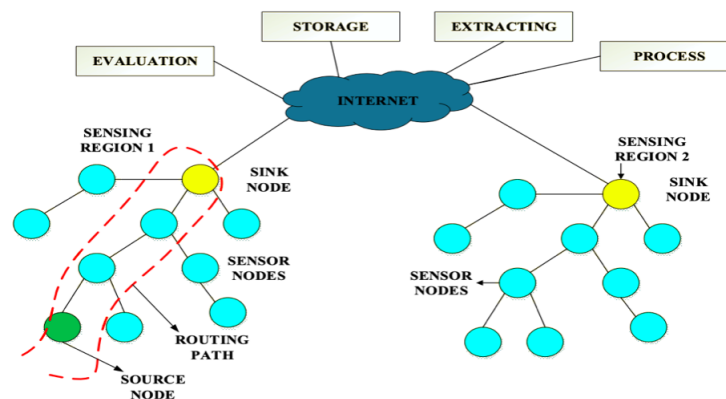


Fig.1.The general structure of wireless sensor networks

Wireless network challenges encompass global IP addressing, fixed nodes, and the availability of energy and capacity in a node [1]. Above mentioned difficulties can be overcome by selecting an appropriate routing algorithm. [2] The general structure of a wireless sensor network is depicted in fig 1. The sink node is linked to the various sensor nodes. One sensing region is formed by combining sensor nodes and sink nodes. The information is transmitted from the source node to the destination node. The path chosen to transfer data from source to destination is called a routing path. Selecting an efficient routing path increases the efficiency of a network and minimizes packet loss. [3][4]. The routing protocol is classified based on the network's architecture and operations and the routing process. The routing protocol's architecture is further defined as flat, hierarchical, and location-specific. In a flat structure, there is no preference for node, and all nodes are treated the same for transmitting data, whereas, in a hierarchy, nodes with higher energy have higher priority for transmission, and nodes with lower energy have the highest priority for receiving data packets [5].[6] In a location-specific routing structure, the sensor nodes are addressed based on their location on location and are updated using the Global Positioning System.

Routing protocols improve the delivery rate of a packet in a network. The routing protocol must be enhanced because nodes are limited in terms of energy and storage, and computing capacity. Reinforcement Learning (RL) [7] [8] is a machine learning technique that is used to efficiently schedule sensor nodes because it does not demand pre-existing environment modeling. Positive and negative reinforcement are different sorts of reinforcement learning. Positive RL refers to events that result from positive behavior, while negative RL refers to events that result from adverse outcomes. As a result, using RL to optimize routing protocols reduces packet delay. For optimization of the network SMO algorithm is utilized. The SMO algorithm [9] [10] follows the FFS structure. The two fundamental concepts of swarm intelligence are self-organization and labor reformatting. Finding Euclidean distances between potential solutions are used to compute the update functions. The primary goal of this paper is to optimize the routing protocol by implementing RL with the SMO algorithm to improve performance and minimize packet loss. The significant contributions of this paper are as follows:

- Propose a new and generic Adaptive Reinforced Learning based Spider Monkey Optimization (ARL-SMO) to determine the optimum path between the source sensor node and the sink node.

The proposed ARL-SMO is statistical all the time by running Rewards and Q-values. Also, it can adaptively change the environment, which means modifying the network for unexpected changes.

- The proposed ARL-SMO is statistical all the time by running Rewards and Q-values. Also, it can adaptively change the environment, which means modifying the network for unexpected changes. Finally, reconfiguring the setting depends upon the network.
- Q-Routing by Boyan, Justin, and Michael Littman(1993) Selecting the next forwarder based on historical learning and current estimation information. Considering the minimal delivery time to learn the best paths
- Different simulation results show that the proposed ARL-SMO can outperform the traditional reinforcement learning and optimization approaches in terms of End-to-End Delay, Communication Overhead, Throughput, Energy Consumption, Network Lifetime, and Packet Loss Rate.

The remainder of the paper is structured as follows: Section 2 highlights the most recent advancements in routing protocol enhancement. Section 3 discusses the approach of the planned effort. The performance of the experimental findings and a comparison of the suggested optimized routing algorithm to the traditional routing protocol. Section 4 summarises the paper's findings and offers recommendations for future network efficiency improvements.

2. Literature Review

The traditional approaches for optimizing routing protocols are addressed in this section. Some of the standard algorithm's and advanced approaches' drawbacks are emphasized. The authors of [11] use an ant-colony optimization approach to improve the routing protocol in a WSN. The transmission of a packet from source to destination saves time and energy. This automatically increases the lifetime of a network. The sending of neighbor cluster heads based on available power reduces network overhead. This method suffers from choosing the shortest path, which consumes the least energy.

An optimized routing protocol by implementing Energy Efficient Clustering and Shortest-Path Routing Protocol is presented in [12]. This algorithm tends to improve:

- The lifetime of the network.
- Optimal usage of battery.
- Reduction of network overhead.
- No packet loss during transmission.

Authors in [13] implement reinforcement learning for the scheduling of sensor nodes. The proper scheduling of sensor nodes avoids Loss of connection in WSN. Reinforcement learning is employed because it does need a prior implementation setup. Nash Q-Learning-based node scheduling algorithm is implemented to maintain a network's coverage. In this algorithm, each node can able to learn its state and its coverage area, and this tends to increase the coverage area in WSN. It also eliminates the coverage duplication of a node and reduces the energy consumption of a node. Since increased consumption of energy by a node decreases the efficiency of a network. Compared to the traditional algorithm, this algorithm tends to improve coverage and connectivity maintenance. Scheduling of node and removal of redundancy node is also performed. This algorithm needs improvement in convergence time and global optimality.

Authors in [14] proposed an improved routing protocol based on reinforcement learning. The network's lifetime and energy consumption tend to improve by utilizing reinforcement learning. The advantage of using RL is that it does not require any prior knowledge of topology to be configured. The nodes will be aware of their energy and will be configured accordingly. Distance between source and destination, available node energy, and the number of hops between source and destination are all factors to consider when implementing a routing protocol.

Various optimization techniques are proposed for Routing in WSN. In WSN, energy consumption is reduced, and also network lifetime is improved in Routing, which is a mandatory

requirement for sensor networks. For example, in case of any sensor is not able to send packets from the source node, it may be affected by the energy drain or power loss issues. This survey highlights the various routing challenges in WSN, which are energy consumption, load balancing, node deployment, data gathering, latency, fault tolerance, and scalability. Several optimization techniques, such as particle swarm optimization (PSO), ant colony optimization (ACO), genetic algorithm (GA), artificial bee colony (ABC), bacterial foraging optimization (BFO), etc. These protocols are established to minimize power consumption which improves the network's lifetime. Most Routing protocols are assumed to have overall information about the network. The network parameters are optimized utilizing the termination time. Fault tolerance and scalability are the key elements to consider in optimization problem-solving [15].

Similarly, authors in [16] have compared the work with several Routing algorithms, and the comparison between these protocols for Routing is elaborated. However, the proposed routing protocol depends upon the network organization (flat architecture, hierarchical architecture, and location architecture), Routing discovery (reactive, proactive, and hybrid), and protocol operation (negotiation-based, multipath-based, query-based, QoS based, and coherent-based). Here the frequently used routing metrics are energy, quality of service, robustness, and minimum hope. The QoS-based routing protocol is designed for sensor network communication. A variant of the routing algorithm is proposed, known as greedy real-time power aware routing (GRPAR), which guarantees to achieve the required QoS. This protocol follows three kinds of policies as follows. (1). A greedy approach is presented for evaluating the transmission path from the source to the destination, (2). The standard transmission protocol is designed for transmission power adaptation, and (3). For packet routing, new neighbors are elected. The simulation results proved that the performance of the proposed work reduces energy consumption and delay and increases the packet transmission and delivery ratio [17].

The fish swarm optimization technique is proposed for Routing in sensor networks. This routing protocol supports consuming the lesser energy for WSNs. Further, this is compared with the existing routing protocol that uses traditional optimization techniques. Experiments were conducted for simulation using OPNET 11.5 simulator that compared with the ERA routing protocol. Finally, the performance is evaluated for several performance metrics, such as End-to-End Delay, Media Access Delay, Throughput, Probability of Successful Transmission from the Source to the Destination, and Signal to Noise Ratio (SNR) [18].

Proposed research objective: Following are the research objectives:

- (i) To Identify strategies for increasing the lifetime of the network by Routing algorithm, which uses a machine learning technique
- (ii) To Minimise energy consumption in the node by reducing the distance of data transmission.
- (iii) To achieve optimal routing path for data transmission and inter-cluster communication.
- (iv) To reduce the hot spot problems by the mobile sink and to address the heterogeneity platform.

Objective 1: To Identify strategies for increasing the network's lifetime by Routing algorithm, which uses machine learning techniques. The packet routing scheme in Wireless Sensor Networks uses various machine-learning techniques. Here the proposed routing strategy is performed with adaptive reinforcement learning with a spider monkey optimization algorithm. The data packets are transmitted in the network using optimal route establishment. This helps to reduce the energy by updating only the Q-value at each instance. This provides adaptive mechanisms to exhibit intelligent behavior in complex and dynamic environments like WSNs. Inspired by such an idea, some intelligent routing protocols have recently been designed for WSNs. The multiple routing paths create frequent channel-switching events due to user interference and overall network performance. The routing parameters such as bandwidth, end-to-end delay, throughput,

overhead, and packet delivery ratio are obtained. By comparing with the other machine learning techniques, the performance of WSN is improved and obtained better performance results.

Proposed ARL-SMO Scheme

3. System Model

The data sensed by sensor nodes for the environment is forwarded to the sink by optimum path, which is the requirement of this paper. Fig.2. describes the overall system model.

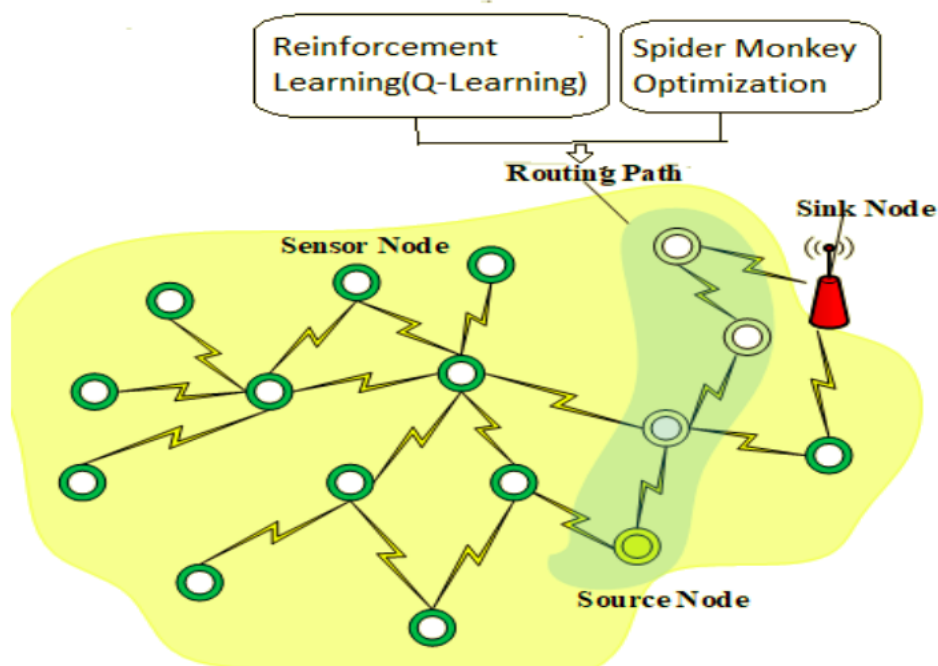


Fig.2. Architecture for ARL-SMO

In ARL-SMO, some of the assumptions are taken into consideration as follows.

- (i) All sensors positioned in the environment are homogeneous such as the sensor's initial energy, communication ability, and computing power must be the same
- (ii) All sensors are restricted for energy, and they cannot be recharged
- (iii) All sensors are distributed randomly throughout the network
- (iv) All sensors' distances can be adjustable but not movable.
- (v) Sink periodically carries the data gathering, and all sensors send packets through optimized route.

In the following sub-sections, the overview of the proposed algorithms is described.

3.1. Spider Monkey Optimization

The spider monkey optimization is considered an effective Swarm Intelligent technique. The SMO algorithm integrates the method of finding food for the spider monkeys. Characteristics of Spider monkeys are explained as the fission-fusion social network. They exist in groups, and each group has several individuals. To resolve the competition issue among the group members, they are further divided into several groups, and the parent group global leader controls the entire group and subgroups in searching for food. Also, the subgroups are controlled by the local leaders who are in subgroups. Depending on the food availability and the territory boundaries

to search, the individuals of each subgroup communicate within and outside the subgroups. The SMO process is divided into four phases considering the FFSS – Fission Fusion Social Structure of the spider monkeys.

- Phase 1 – This group checks for the classification of food and initiates consuming the food.
- Phase 2 – A practical review of the exact sites of the food collection and the food distribution is carried.
- Phase 3 – Here, the Local forerunner restores his best standing in the group. The entire group reviews the diet because of his lack of renewal of his best position.
- Phase 4 – An excellent position is reopened by the international pioneer. The local pioneer limit and the global pioneer limit give false reviews to make independent decisions. Fig 4 represents the flow process of the Spider Monkey Optimization Algorithm.
 - Firstly, the initialization process is carried out wherein the control parameters are defined, the population is generated randomly, and the leaders are assigned to the groups and the subgroups as the global leader and the local leaders, respectively.

$$S_{ij} = S_{jlow} + U(0,1) * (S_{jhigh} - S_{jlow})$$

The S_{jlow} and S_{jhigh} are the boundary limits of the Spider Monkey i in the j vector. $U(0,1)$ is the arbitrary numbers 0 and 1.

- Second is the computation of the fitness solution. The best route to reach food must be determined upon the initialization process. For this purpose, the fitness of the Routing is calculated such that the route has the shortest distance and fewer hops.
- The third is the local leader phase, and the global Fourth is the regional leader decision phase and global leader decision phase

3.2. ARL

The Adaptive reinforcement learning approach is considered an effective artificial Intelligent technique. The ARL approach is applied to resolve the network lifetime issue. In this approach, the sensor node activities are priorly scheduled and maintained. For this process, at least one sensor node path route is required to carry the source sensor node information. And this path route is selected by the Spider Monkey Optimization algorithm. In the ARL approach, each sensor node is considered an agent node. It communicates with the environment based on the feedback trial and error method to study the optimum technique by increasing the growing rewards. Fig 5 represents the flow process of standard reinforcement learning.

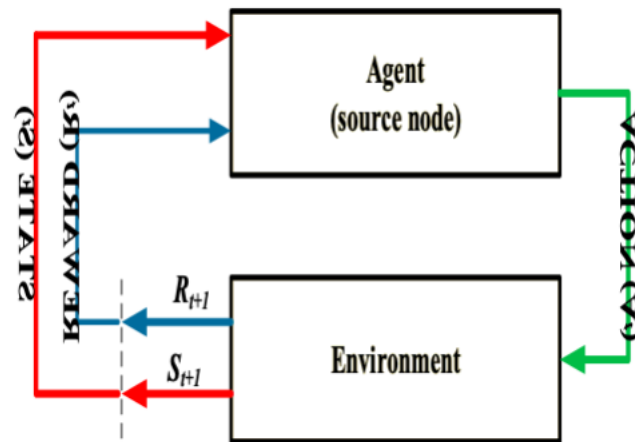


Fig 3 Flow process of standard reinforcement learning

The elements involved in this process are the environment, the action, the state, and the reward. Any particular agent initializes the operation. The agent node initiates an action corresponding to the status of the state via the strategical function. The agent executes the effort to enter the next state. Meanwhile, the network returns the R-value for rewarding or punishing the action.

3.3. ARL-SMO

Adaptive Reinforcement Learning based Spider Monkey Optimization Algorithm (ARLSMO) involves four primary elements. The agent is the Spider Monkey (SM) in the population, and the environment is the Wireless Sensor Network, as given in this paper. The State "S" is the position of each SM in the population "X," and the Action "A" is the velocity "v" of the strategic prediction represented by the strategic function ". The reward value "R" depends on the fitness value function "f" of the optimization issue. Thus, to determine the SM prediction velocity, the strategic function "" is estimated based on the resultant reward value "R." The fitness is computed in SMO for Routing path selection as follows.

The residual energy of the sensor node is computed for every data transmission. The mathematical formula for the residual energy computation is given below.

$$R_{E(i)} = E_I - E_C [1,eq.(1)]$$

Where E_I is the initial energy of the sensor node, represents the energy consumption for sensor nodes and, thus, the mean value of residual energy is given below.

$$MR_{E(i)} = \sum_{i=1}^{i=n} \frac{E_I - E_C}{N} [1,eq.(2)]$$

Distances between two sensors are computed by their current coordinate positions through Euclidean Distance, which can be expressed mathematically follows.

$$D = \sqrt{(Y_1 - Y_0)^2 + (X_1 - X_0)^2} \quad [1,eq.(3)]$$

In ARLSMO, the SM agent forecasts the velocity concerning the strategic function ”.”

$$Velocity(SM + 1) = \mu [X(SM)][1,eq.(4)]$$

Then the network model was constructed, trained, and evaluated to discover the shortest path route for effective communication with an improved network lifetime.

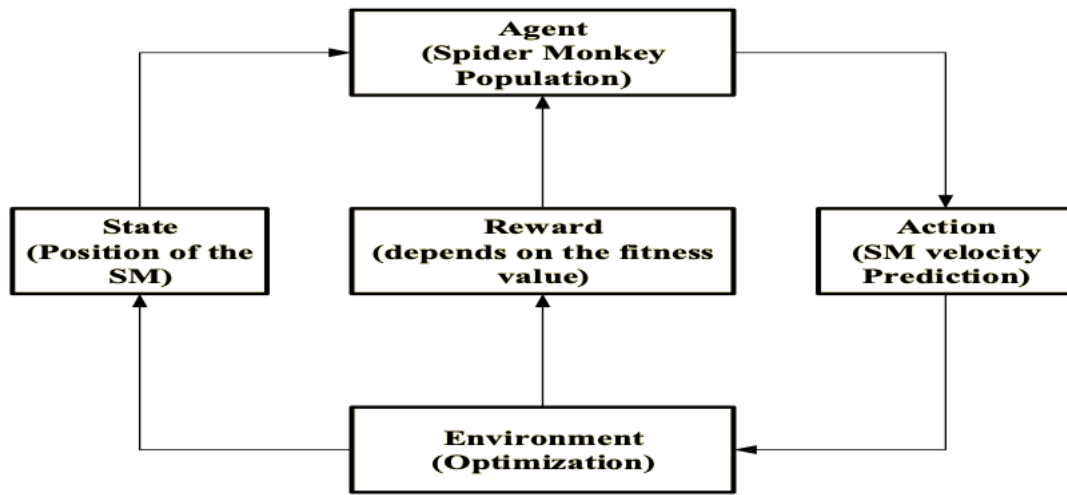


Fig.4. Flow process of proposed ARLSMO technique

4. Experimental Setup

In this section, the performance of the proposed ARL-SMO scheme is implemented utilizing a network simulation tool.

4.1. Simulation Parameters

This section briefs the experimentation of the proposed ARLSMO model on the simulation platform. The simulation tool used for this study was the Network Simulator tool - NS 2.34 version. The simulation setup involves the WSN nodes – 100 No.s placed in 1000 meters x 1000 meters square region for about 36 milliseconds. Each WSN node goes randomly along with the network at various speed ranges. Every node has a similar degree of transmission, around 250 meters. CBR - Constant Bit Rate delivers a constant speed of the data packet transmission on the network to limit the traffic rate. Table 1 gives the simulation setup parameters used.

Number of Nodes	100	Source of traffic	CBR
Area dimension	1000 X 1000	Packet dimension	512 bytes
MAC address	802.11g	Mobility Type	Random Way Point
Radio limit	250m	Radio propagation model	2 Ray Ground Reflection model
Simulation speed	36ms	Data rate	250kbps
Antenna type	Omnidirectional	Radius	40m
MAC type	IEEE 802.15.4	Simulation Time	100s
Radiofrequency	2.4GHz	Channel type	Wireless / Channel
Queue type	Priority queue	Control packet size	32bytes
Data packet size	1024bytes	Energy initial	3J

Table 1. Simulation Setup of ARLSMO implementation

4.2. Comparison Analysis

The proposed model Adaptive Reinforcement Learning based Spider Monkey Optimization (ARLSMO) method is used to provide interference-free communication more efficiently than the existing MOSMO[19] Flock-CC [20]. The ARLSMO approach was utilized for providing scheduled pack delivery enabling energy-efficient multiple-path Routing that retrieves the previous routing information of the entire WSN network. The spider monkey algorithm selects the higher resource limit and the guaranteed node for the routing process. This approach increases the network lifetime, thereby reducing the packet loss rate. The performance of the proposed routing algorithm – Spider Monkey Optimization, along with the effective machine learning technique – Adaptive Reinforcement learning, is assessed. The results for the end-end delay, the power consumption, the throughput, the overhead, and the packet delivery ratio were simulated. The simulation results of the proposed model ARLSMO compared with the MOSMO and Flock-CC for analyzing the performance.

4.2.1. End-to-End Delay Analysis

Figure 5 depicts the end-end delay result for the proposed ARLSMO model considering the time consumption range for transmitting the routed packets from the source node to the destination node. The complete detail about the individual nodes is contained in the routing data table.

$$\text{End - EndDelay} = \text{EndTime} - \text{StartTime} [2, eq. (5)]$$

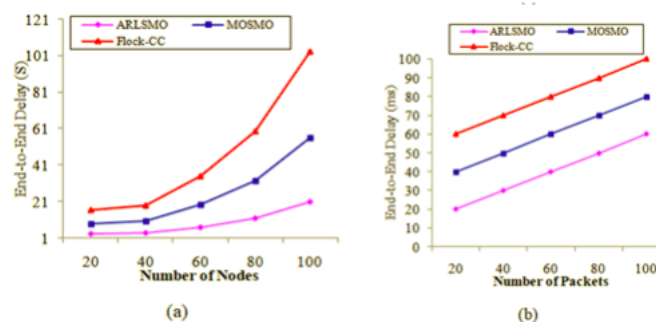


Figure 5 Comparison plot for End-End Delay analysis (a). Number of Nodes and (b). Number of Packets

The end-end delay factor was reduced in the proposed ARLSMO model than in the other existing models like the MOSMO and Flock-CC. The results suggested that the proposed method showed a 35% decreased end-end delay compared to the MOSMO model.

4.2.2. Communication Overhead Analysis

Figure 6 depicts the comparison plot for the communication overhead of the proposed ARLSMO model with other existing models. The ARLSMO algorithm was utilized for providing scheduled pack delivery enabling energy-efficient multiple-path Routing that retrieves the previous routing information of the entire WSN network. This data was efficiently forwarded to a particular target node only. Therefore, very little energy was consumed, and there was no communication overhead.

$$\text{Communication overhead} = \left(\text{No. of Packet} \frac{\text{Losses}}{\text{Received}} \right) * 100 [2,eq. (6)]$$

The communication overhead factor was reduced in the proposed ARLSMO model than in the other existing models like the MOSMO and Flock-CC. The results suggested that the proposed method showed a 35% decreased communication overhead compared to the MOSMO model.

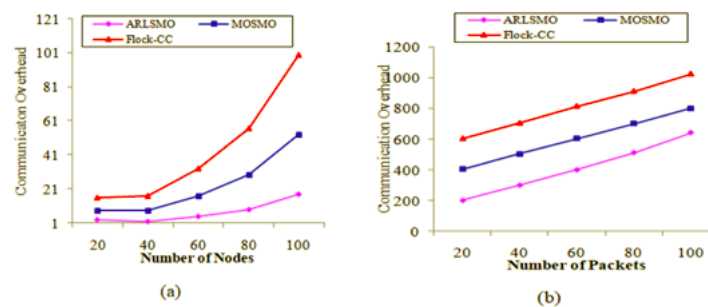


Figure 6 Comparison plot for Communication Overhead Analysis (a). Number of Nodes and (b). the number of Packets

4.2.3. Throughput Analysis

Figure 7 depicts the comparison plot of the throughput of the proposed ARLSMO model with other existing models. It is the number of received packets from the number of packets transmitted at a specific speed limit. The velocity of the source nodes is not stable, but the simulation mobility was fixed at 100 bits per second. The SMO algorithm was utilized to opt lesser counter-value node for the routing process. In the proposed ARLSMO model, the throughput. The rate was raised than that of the other existing methods, MOSMO and Flock-CC. The mathematical expression for throughput is given below.

$$\text{Throughput} = (\text{Number of packet received/Sent}) * \text{speed} [2,eq. (7)]$$

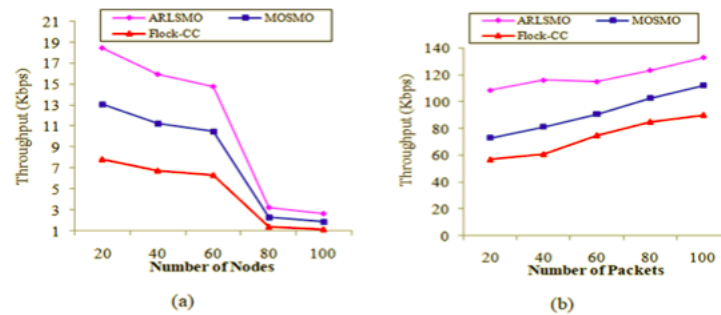


Figure 7 Comparison plot for Throughput Analysis (a). Number of Nodes and (b). Number of Packets

4.2.4. Network Lifetime Analysis

Figure 8 depicts the comparison plot for the proposed ARLSMO with other existing models. The network lifetime was estimated with the nodes involved in the WSN network's processing time.

$$\text{Network Lifetime} = \text{timetaken to utilize network} / \text{overallability} [2,eq(8)]$$

In the proposed ARLSMO approach, the link connectivity was acquired, and thus the network's lifetime was improved than the other existing models like MOSMO, and Flock-CC model.

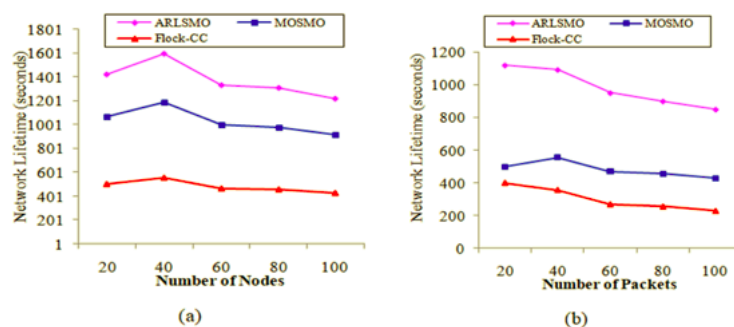


Figure 8 Comparison plot for Network Lifetime Analysis (a). Number of Nodes and (b). Number of Packets

In WSN, routing protocol plays an essential role in determining the performance of the entire network. Generally, in the WSN network, the routing protocol is used for communication purposes, wherein the data packets are transmitted between the sensor nodes and the Base station or the sink nodes. The issue is the higher energy consumption due to updating the Quality value of each node to all nodes on the network, which demands high bandwidth and Quality of Services (QoS) provisioning. The Spider Monkey routing method is used to address these challenges. Furthermore, coverage and connection must be maintained, and the adaptive reinforcement learning technique is used to do this. This proposed technology will be energy-efficient for increasing network lifetime and communication effectiveness. Fig 9 depicts the comparison plot for the energy consumption analysis of the proposed model with other existing models. The extra energy utilized for the data transmission over the WSN network is the estimation of the energy consumption initializing the energy level to stopping the energy level.

In the proposed ARLSMO method, energy consumption was reduced to an extent where there occurs less communication overhead than that of the other existing methods, MOSMO and Flock-CC.

$$\text{EnergyConsumption} = \text{InitialEnergy} - \text{FinalEnergy} [2,eq.(9)]$$

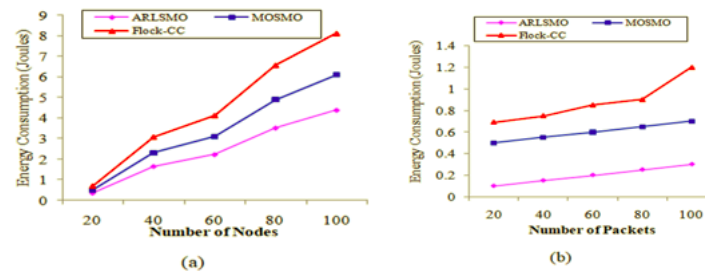


Figure 9 Comparison Plot for Energy Consumption Analysis (a). Number of Nodes and (b). The number of Packets

4.2.5. Packet Loss Rate Analysis

Figure 10 depicts the comparison plot for the Loss of packets analysis of the proposed model with the existing models.

$$\text{Packetloss} = \left(\text{Number of packet} \frac{\text{dropped}}{\text{Sent Packets}} \right) * 100 [2,eq.(10)]$$

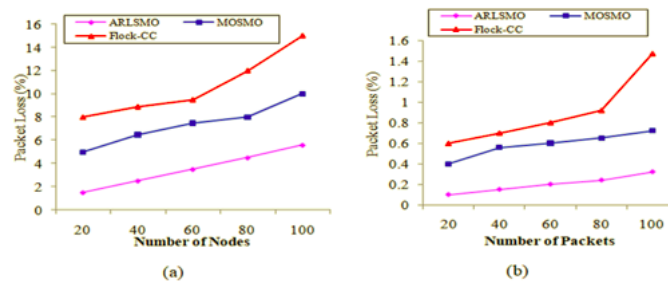


Figure 10 Comparison plot for Packet Loss Analysis (a). Number of Nodes and (b). Number of Packets

During the transmission process, packet loss occurs, and it is determined by the node's packet loss upon the weakest connectivity to estimate the traffic-free data transmission. The results indicated that the proposed model showed excellent energy efficiency without losing packets during transmission. Thus the proposed model outperformed the other existing models, MOSMO and the Flock-CC.

5. Conclusion

The performance of the proposed routing algorithm – Spider Monkey Optimization, is implemented along with the effective machine learning technique – Adaptive Reinforcement learning, is assessed by experimenting over the Wireless Sensor Network using the Network Simulator – NS module 2.32. After comparing the proposed model with the MOSMO model and the Flock CC model, the proposed model outperformed in all the metrics, including the end-to-end delay, throughput rate, communication overhead, packet delivery ratio, etc.

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